

Database Localization Plan and Implementation Report

1. Introduction

Localization is the process of adapting a system to support multiple languages and cultural contexts. In the ClassIQ system, localization is important because users may use the application in different languages such as English, Sinhala, and Arabic. Therefore, both the user interface and database content are designed to support multiple languages.

2. Objectives

- Support multiple languages (English, Sinhala, Arabic)
- Store multilingual data efficiently in the database
- Enable dynamic language switching
- Maintain consistency between UI and database
- Allow easy expansion for future languages

3. Localization Strategy

3.1 UI Localization:

The UI uses resource bundle files such as `messages_en.properties`, `messages_si.properties`, and `messages_ar.properties`.

3.2 Database Localization:

Separate translation tables are used to store multilingual data, linked using IDs and language codes.

3.3 Language Selection:

The selected language is stored in the session and applied across the system.

4. Database Design for Localization

Translation tables such as **student_translation**, **teacher_translation**, **subject_translation**, **gradecategory_translation**, **role_translation**, and **marksheet_translation** are used.

Each table contains:

- Reference ID
- Language code
- Translated values

This allows storing multiple language versions for the same data.

5. Implementation Details

- Language is stored in session
- Queries retrieve data based on selected language
- Translation tables are joined with main tables
- English is used as fallback when translation is missing
- RTL layout is applied for Arabic

6. Challenges Faced

- Managing multiple translation tables
- Handling missing translations
- Maintaining UI consistency
- Implementing RTL layout
- Complex database queries

7. Solutions

- Used structured translation tables
- Added fallback to English
- Ensured consistent naming
- Tested across all languages
- Applied RTL adjustments

8. Testing and Validation

- Tested dynamic language switching
- Verified Arabic RTL layout
- Checked translated data retrieval
- Ensured UI consistency

9. Conclusion

The ClassIQ system successfully implements database localization using translation tables. The design supports scalability, flexibility, and consistent multilingual functionality.